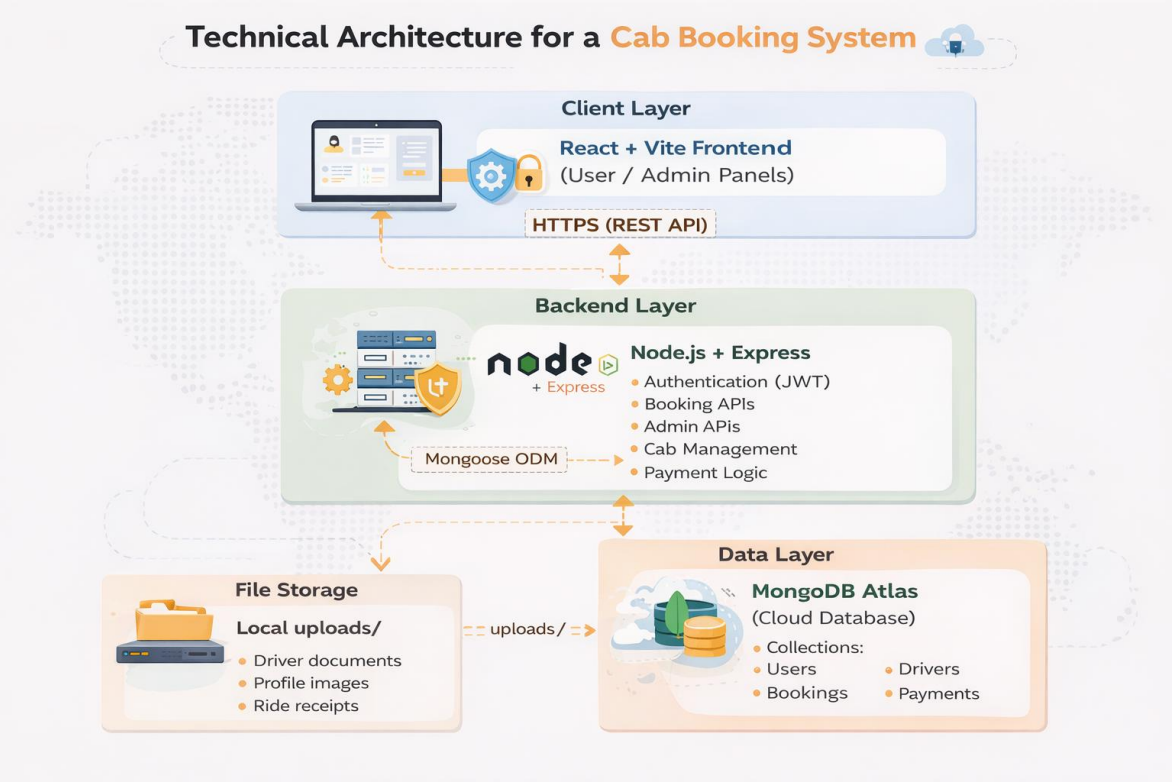
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	8 march 2026
Team ID	
Project Name	Cab Booking
Maximum Marks	4 Marks

Technical Architecture



Technology Stack

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application for users and admin	React (Vite), React Router, CSS (custom theme)
2	State Management	Manage authentication state across components	Context API
3	HTTP Client	Communicate with backend API	Axios (with interceptors for token)
4	Backend Server	REST API for business logic	Node.js, Express.js
5	Authentication	User/admin login, JWT issuance & validation	JWT, bcrypt
6	File Upload	Handle cab image uploads	Multer (disk storage)
7	Database	Store users, admins, cars, bookings	MongoDB (Mongoose ODM)
8	Cloud Deployment	Host application (optional)	(e.g., Render, Vercel, Atlas)
9	Environment Config	Manage environment variables	Dotenv

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology used
1	Open-Source Frameworks	Entire stack is open-source	React, Node.js, Express, MongoDB
2	Security Implementations	Password hashing, JWT tokens, protected routes, input validation	bcrypt, JWT, middleware
3	Scalable Architecture	Stateless API (JWT), can be scaled horizontally; MongoDB can be sharded	MERN stack inherently supports scaling
4	Availability	Can be deployed on multiple servers with load balancer; database replication possible	Cloud services (AWS, etc.)
5	Performance	Lazy loading, code splitting, efficient queries (indexes)	React Suspense, MongoDB indexes